

Screwworm Report Network: an Integrated Biosurveillance System

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I3C: Intern Innovative Idea Challenge



Problem Statement

Since 2023, the deadly **New World Screwworm (NWS)** has spread from Central America to the U.S., endangering livestock by infesting cattle wounds with larvae.^[1] This escalating crisis threatens the \$160 billion cattle industry, risking devastating losses for farmers and higher beef prices for consumers.^[2] Immediate action is essential to protect livestock and the economy.



Current State of the Art and Limitations

Surveillance: About 100 traps are deployed at the U.S.-Mexico border^[3]

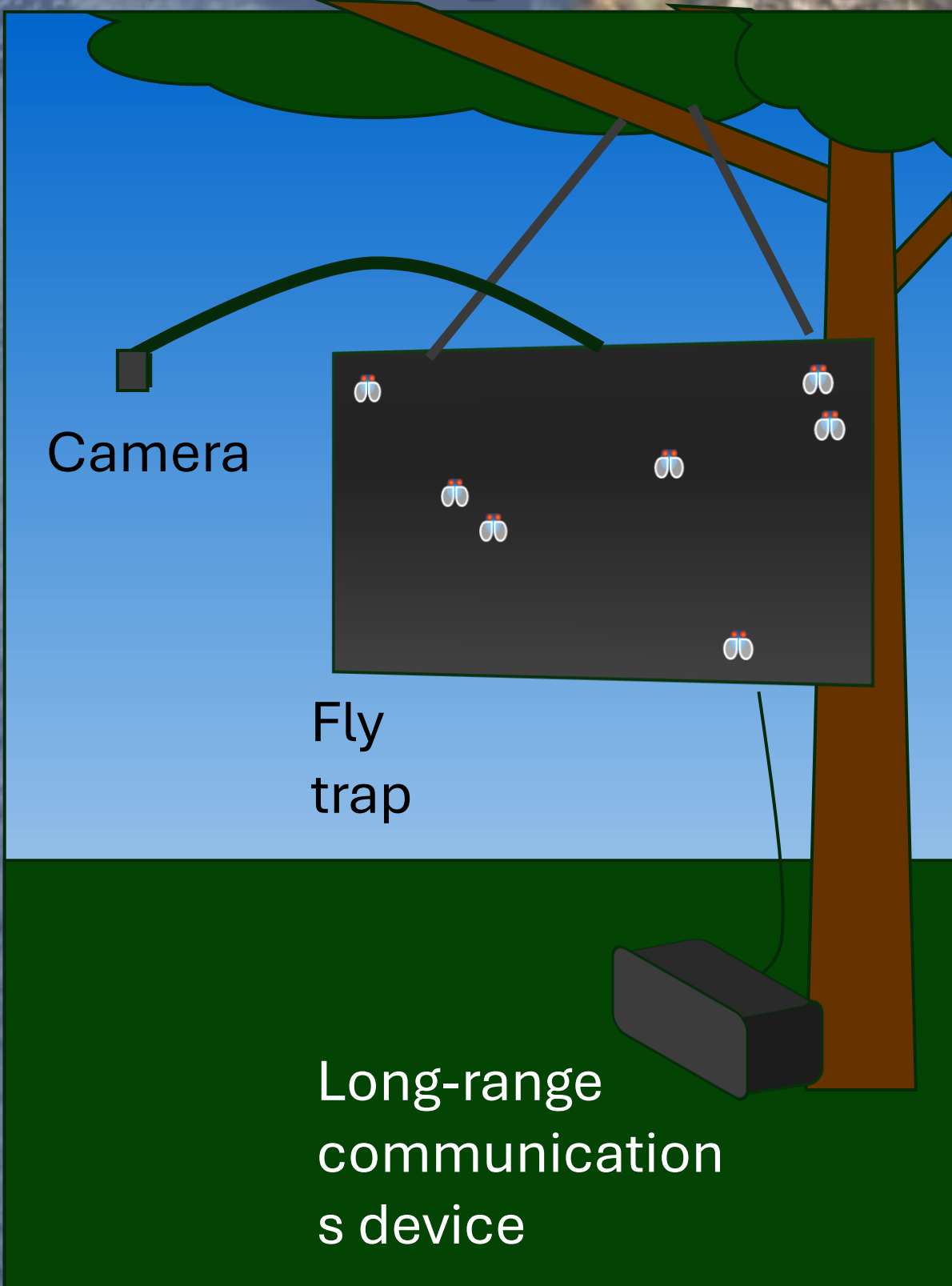
Eradication: Sterile fly release is effective, but limited in supply

Trap monitoring: USDA traps laced with swormlure-4 attractant require manual checks twice weekly

Limitations: Current methods are slow and lack automation

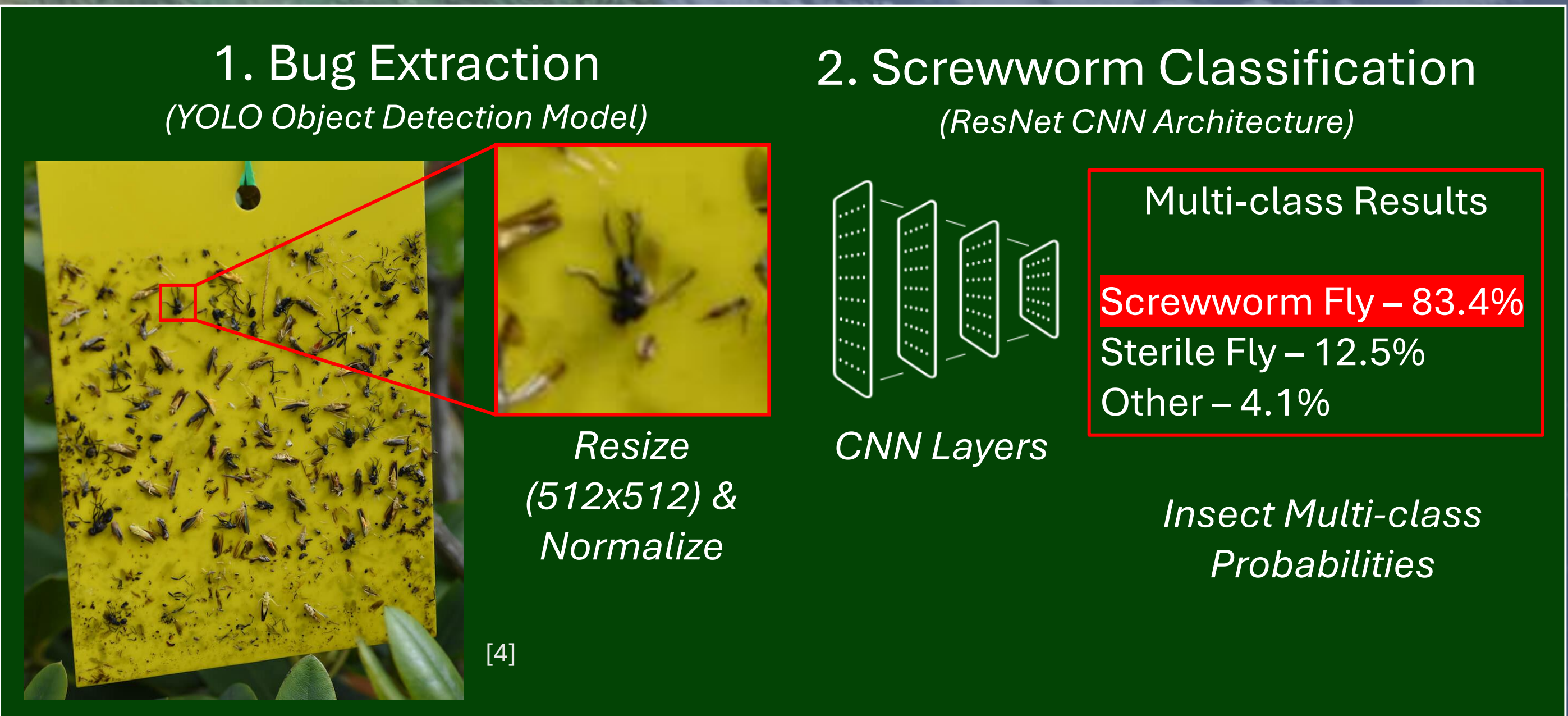


Our Approach



We propose an integrated biosurveillance system:

- Sticky traps with chemical attractant
- Built-in motion sensor camera
- Neural network trained on NWS images
- Automatically populated GIS-powered tracking map
- Forecasting tool for sterile fly dispersal locations



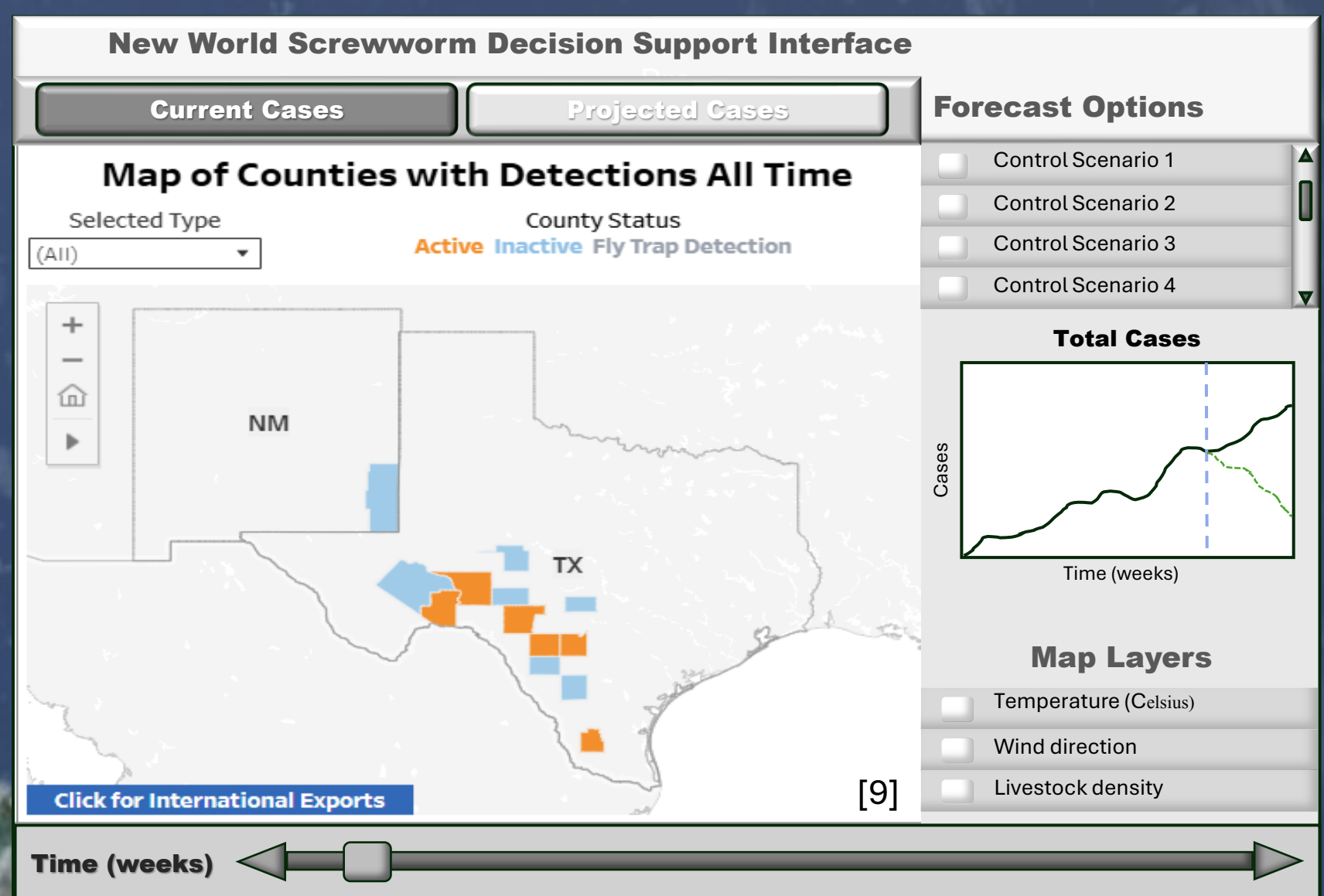
Two-Stage NWS Identification

(1) YOLO vision object detection model extracts individual insects from sticky trap image

(2) Extracted insects are standardized and classified with **ResNet CNN Architecture** to output multi-class species confidence scores

Modeling and Simulation

- **Real-Time GIS Mapping:** Updates with confirmed NWS detections, providing outbreak hotspots to policymakers
- **Dynamic Risk Projection Model:** Uses integrated agent-based modeling to evaluate potential control scenarios.
- **Scalable and Multi-Purpose Design:** Data collection from remote areas and can be expanded to other pests, increasing value long-term.

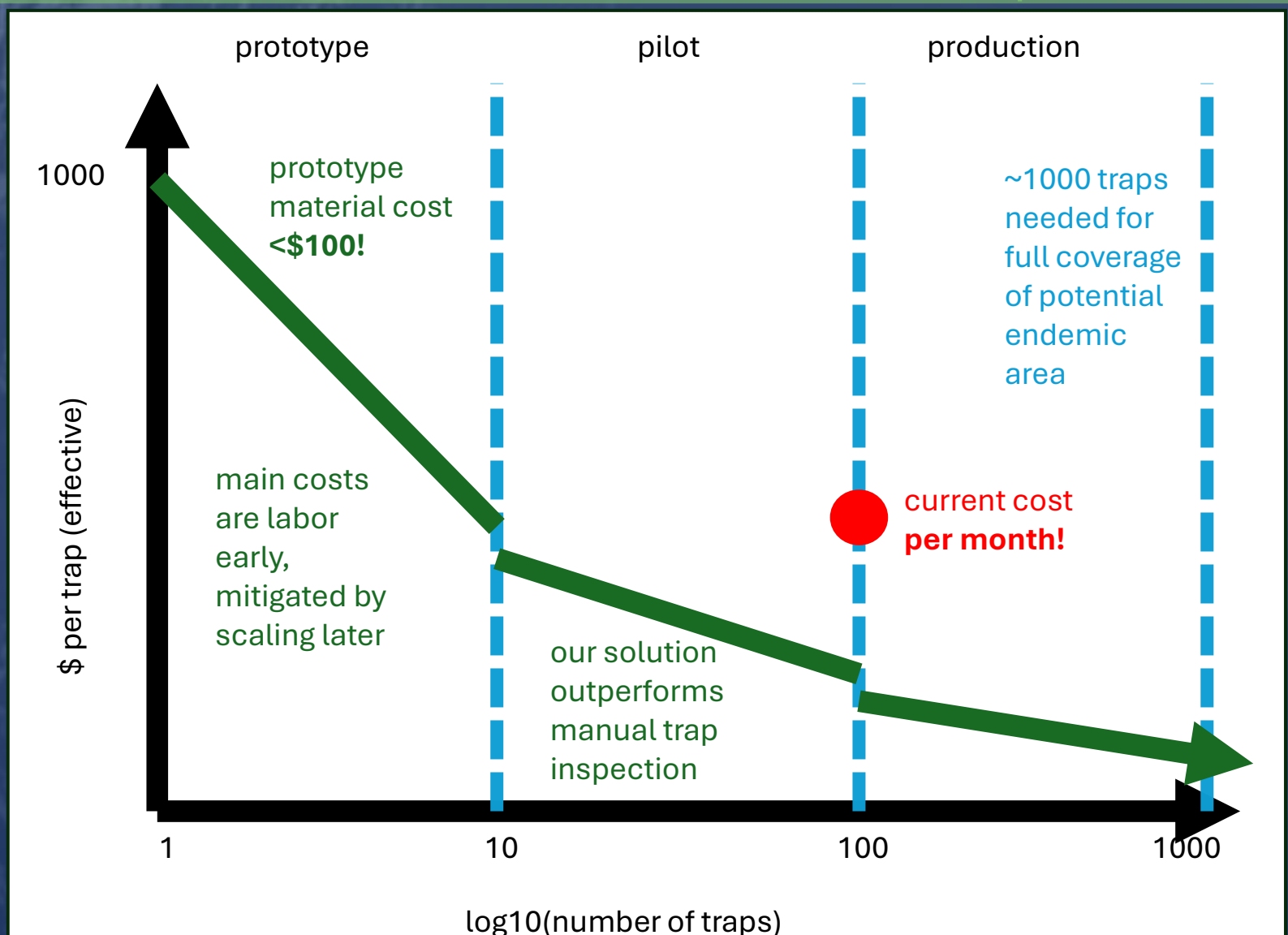


Long Range Communications

- LoRa: cheap radio-based long-range communications
- Networked LoRa avoids cost of cellular/satellite
- Limitations include connection speed and reliability
- Mitigated by modest transmission rate and low uptime

Cost Analysis

- Cost to beef industry potentially catastrophic
- Lower ongoing costs than previous solution
- Main costs are labor, compute, comms



Future Work

- Enhance tracking of fluorescently tagged sterile flies vs non-sterile flies ^[8]
- Refine image classification to reduce confusion with look-alikes
- Propose integrating the system with USDA traps with swormlure-4 attractant or test varying amounts of formulated attractant (projected cost \$530 per liter) ^[7]

Sources

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Conclusion: The proposed Screwworm Report Network transforms future agricultural decision-making, potentially mitigating the outbreak in an efficient, cost-effective way.